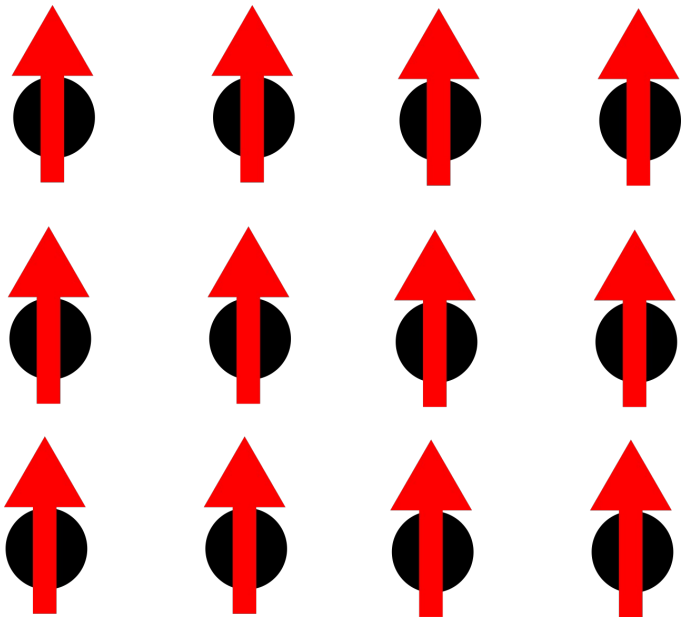
Classical magnetism and magnons

Today’s learning outcomes

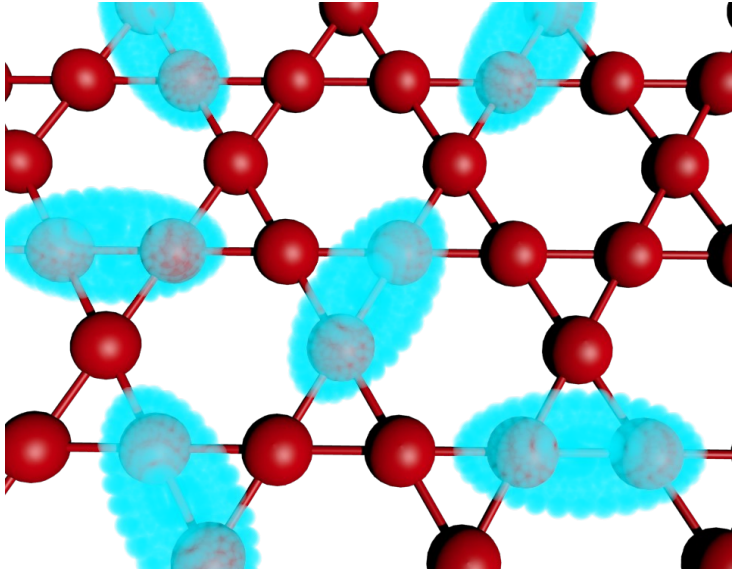
- Derive minimal models for symmetry-broken magnets
- Understand the difference between itinerant and localized magnets
- Identify the quantum excitations of ordered magnets

Classical and quantum magnetism

Ferromagnet & antiferromagnets

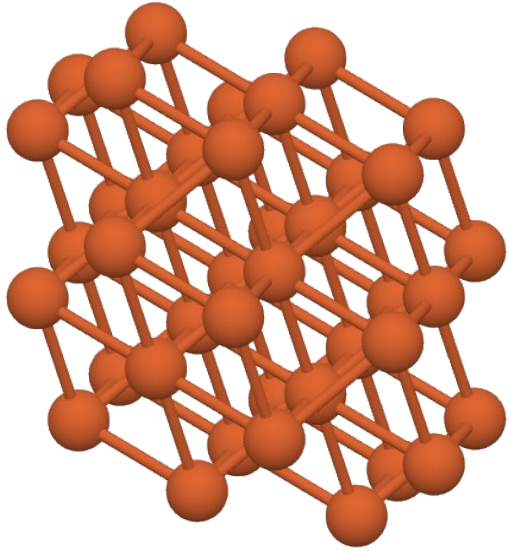


Quantum spin-liquids



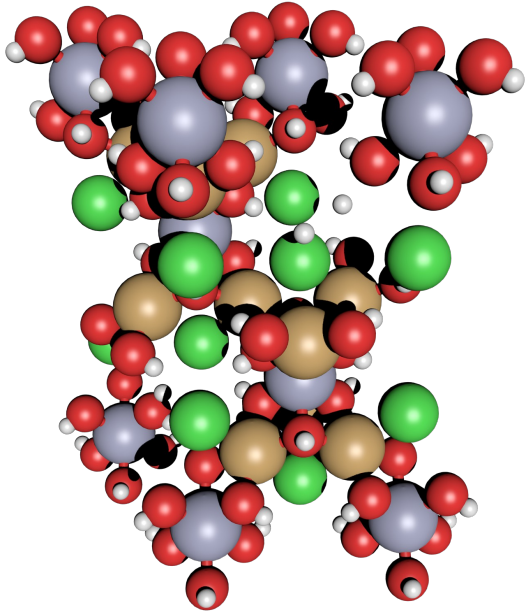
Classical and quantum magnetism

Ferromagnets & antiferromagnets



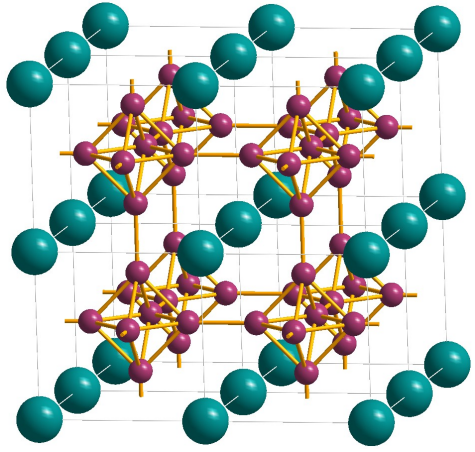
Iron

Quantum spin-liquids



Herbertsmithite

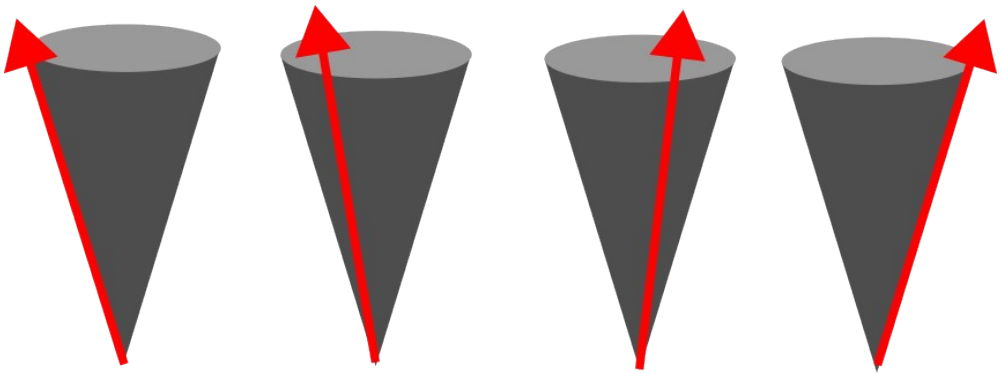
Heavy-fermion insulators



SmB_6

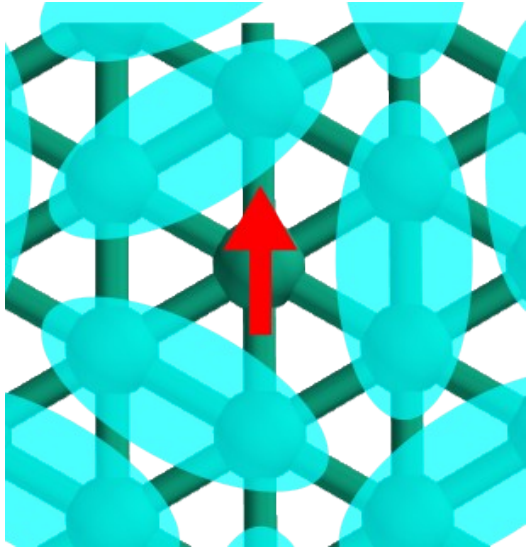
Two excitations in magnetic materials

Magnons



S=1
No charge

Spinons



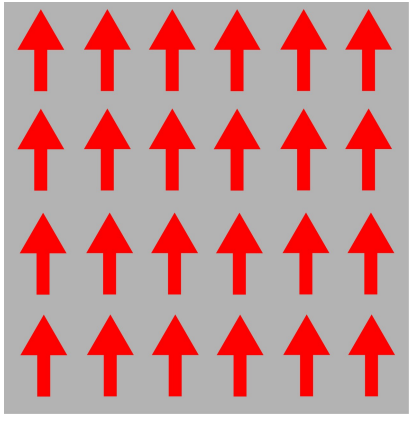
S=1/2
No charge

A reminder from previous sessions

Electronic interactions are responsible for symmetry breaking

Broken time-reversal symmetry

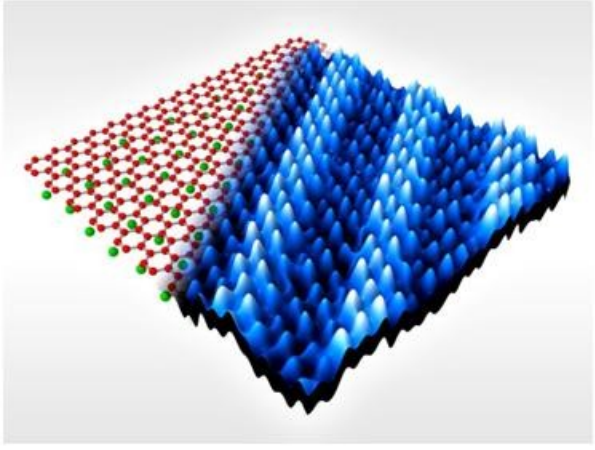
Classical magnets



$$\mathbf{M} \rightarrow -\mathbf{M}$$

Broken crystal symmetry

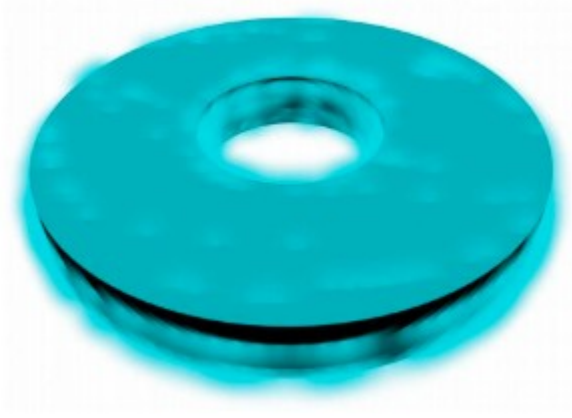
Charge density wave



$$\mathbf{r} \rightarrow \mathbf{r} + \mathbf{R}$$

Broken gauge symmetry

Superconductors



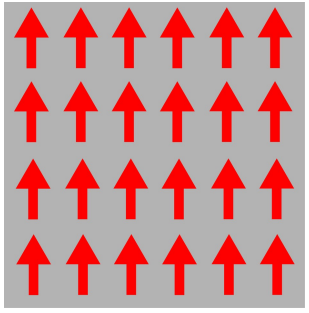
$$\langle c_\uparrow c_\downarrow \rangle \rightarrow e^{i\phi} \langle c_\uparrow c_\downarrow \rangle$$

Correlations and mean field

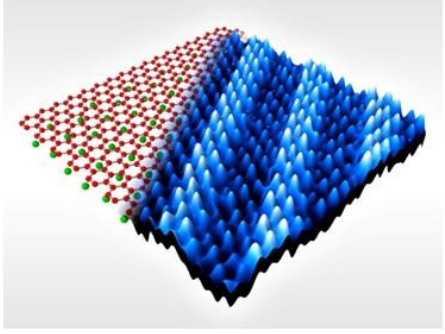
Many quantum states can be approximately described by mean field theories

$$H = \sum_{ij} t_{ij} c_i^\dagger c_j + \sum_{ijkl} V_{ijkl} c_i^\dagger c_j c_k^\dagger c_l$$

Magnets



Charge density waves



Superconductors



Interactions and mean field

Free Hamiltonian ***Interactions***

$$H = \sum_{ij} t_{ij} c_i^\dagger c_j + \sum_{ijkl} V_{ijkl} c_i^\dagger c_j c_k^\dagger c_l$$

What are these interactions coming from?

- Electrostatic (repulsive) interactions
- Mediated by other quasiparticles (phonons, magnons, plasmons,...)

The net effective interaction can be attractive or repulsive

Magnetism is promoted by repulsive interactions

A simple interacting Hamiltonian

Free Hamiltonian ***Interactions (Hubbard term)***

$$H = \sum_{ij} t_{ij} [c^\dagger_{i\uparrow} c_{j\uparrow} + c^\dagger_{i\downarrow} c_{j\downarrow}] + \sum_i U c^\dagger_{i\uparrow} c_{i\uparrow} c^\dagger_{i\downarrow} c_{i\downarrow}$$

What is the ground state of this Hamiltonian?

$U < 0$ Superconductivity

$U > 0$ Magnetism

The mean-field approximation

Mean field: *Approximate four fermions by two fermions times expectation values*

Four fermions (not exactly solvable) ***Two fermions (exactly solvable)***

$$U c^\dagger_{i\uparrow} c_{i\uparrow} c^\dagger_{i\downarrow} c_{i\downarrow} \approx U \langle c^\dagger_{i\uparrow} c_{i\uparrow} \rangle c^\dagger_{i\downarrow} c_{i\downarrow} + ... + h.c.$$

$$U c^\dagger_{i\uparrow} c_{i\uparrow} c^\dagger_{i\downarrow} c_{i\downarrow} \approx M \sigma^z_{ss'} c^\dagger_{i,s} c_{i,s'} + h.c.$$

For $U > 0$
i.e. repulsive interactions

Magnetic order
$$M \sim \langle c^\dagger_{i\uparrow} c_{i\uparrow} \rangle - \langle c^\dagger_{i\downarrow} c_{i\downarrow} \rangle$$

The mean-field approximation

The non-collinear mean-field Hamiltonian

$$U c^\dagger_{n\uparrow} c_{n\uparrow} c^\dagger_{n\downarrow} c_{n\downarrow} \approx M^\alpha_n \sigma^\alpha_{ss'} c^\dagger_{n,s} c_{n,s'} + h.c.$$

Non-collinear magnetic order

$$M^z_n \sim \langle c^\dagger_{n\uparrow} c_{n\uparrow} \rangle - \langle c^\dagger_{n\downarrow} c_{n\downarrow} \rangle$$

$$M^x_n \sim \langle c^\dagger_{n\uparrow} c_{n\downarrow} \rangle + \langle c^\dagger_{n\downarrow} c_{n\uparrow} \rangle$$

$$M^y_n \sim i\langle c^\dagger_{n\uparrow} c_{n\downarrow} \rangle - i\langle c^\dagger_{n\downarrow} c_{n\uparrow} \rangle$$

A Hamiltonian for a weakly correlated magnet

Free Hamiltonian ***Exchange term***

$$H = \sum_{ij} t_{ij} [c^{\dagger}_{i\uparrow} c_{j\uparrow} + c^{\dagger}_{i\downarrow} c_{j\downarrow}] + M \sum_{i} \sigma^{z}_{s,s'} c^{\dagger}_{i,s} c_{i,s'}$$

Here we assume that interactions are weak (in comparison with the kinetic energy)

What if interactions are much stronger than the kinetic energy?

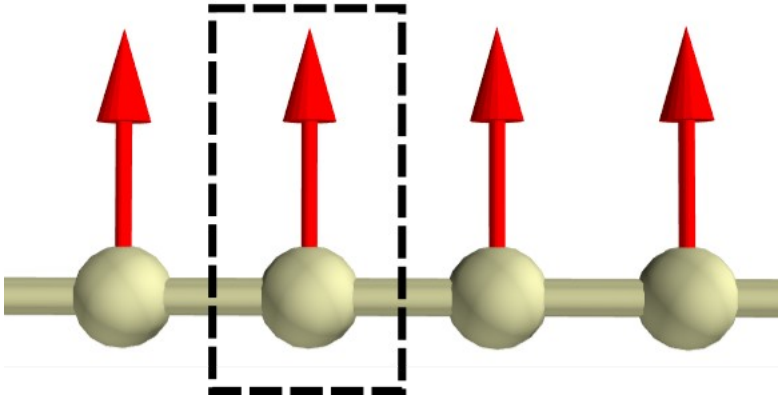
Solving the interacting model at the mean-field level in a 1D chain

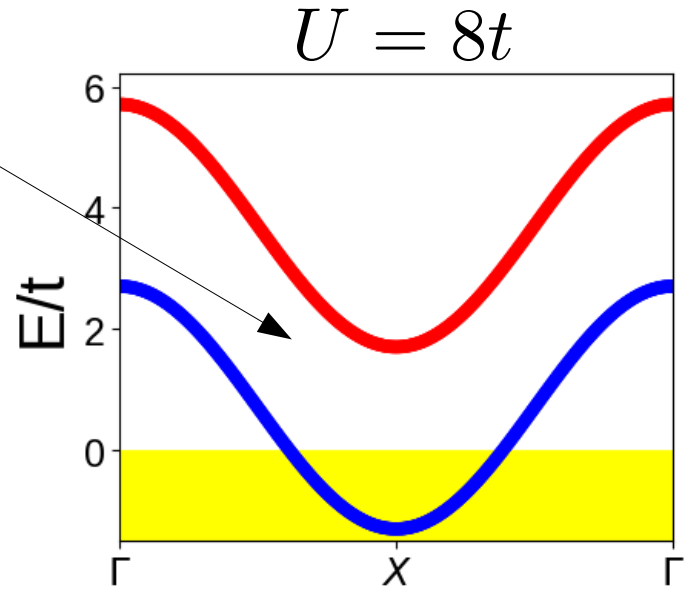
We will take the interacting model and solve it at the mean field level

$$H = \sum_{ij} t_{ij} [c^\dagger_{i\uparrow} c_{j\uparrow} + c^\dagger_{i\downarrow} c_{j\downarrow}] + \sum_i U c^\dagger_{i\uparrow} c_{i\uparrow} c^\dagger_{i\downarrow} c_{i\downarrow}$$

Filling 0.2 (full would be 1)

Interaction-induced splitting

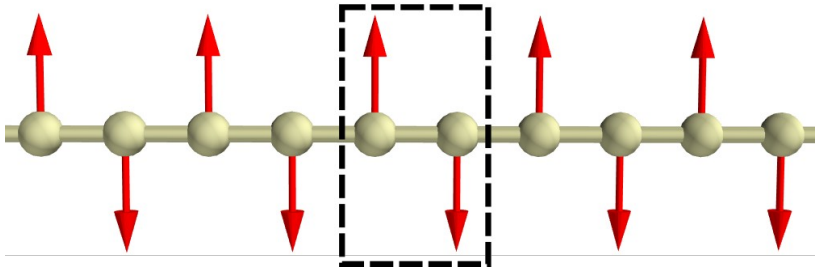


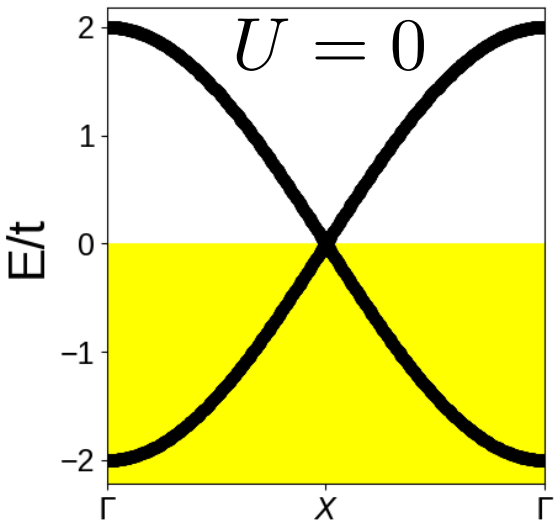


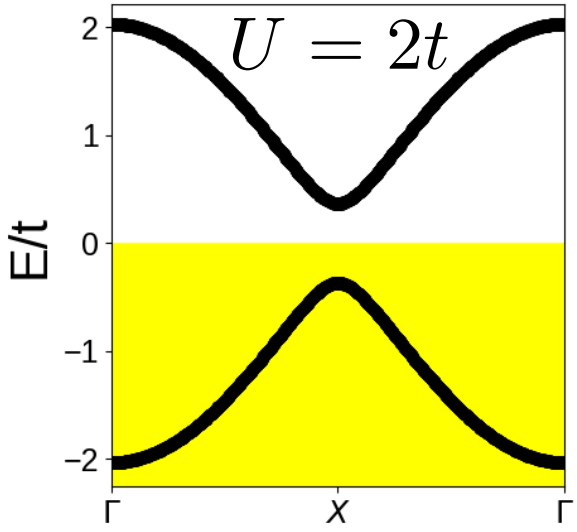
Solving the interacting model at the mean-field level in a 1D chain

Let us do again a 1D, but now with 2 sites per unit cell and at half filling

$$H = \sum_{ij} t_{ij}[c^\dagger_{i\uparrow}c_{j\uparrow} + c^\dagger_{i\downarrow}c_{j\downarrow}] + \sum_i U c^\dagger_{i\uparrow}c_{i\uparrow}c^\dagger_{i\downarrow}c_{i\downarrow}$$

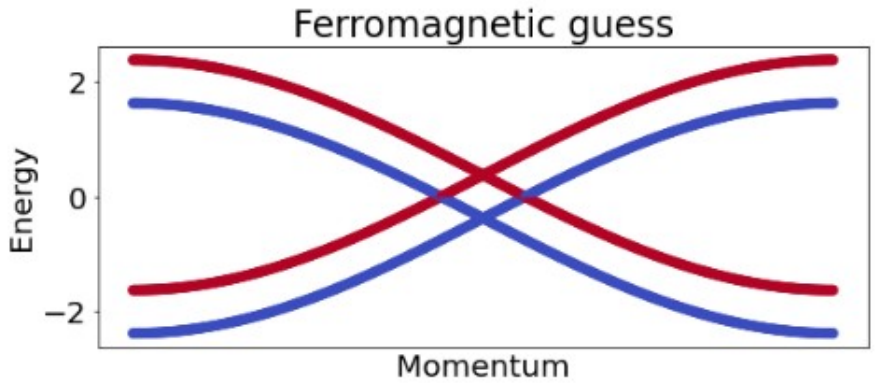


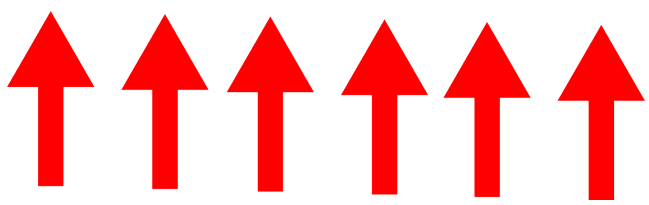


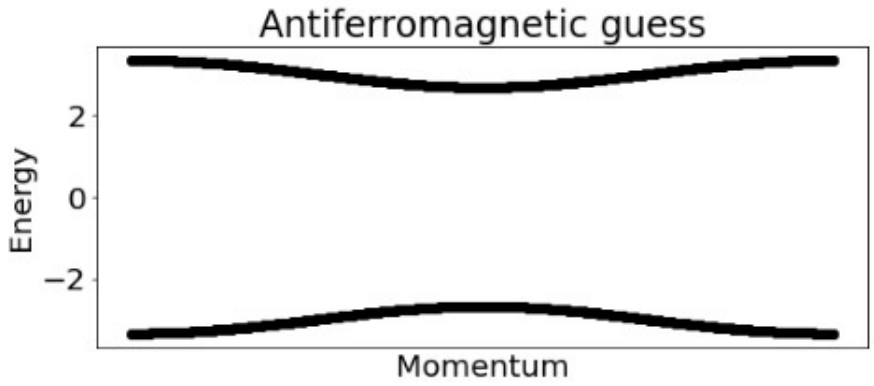


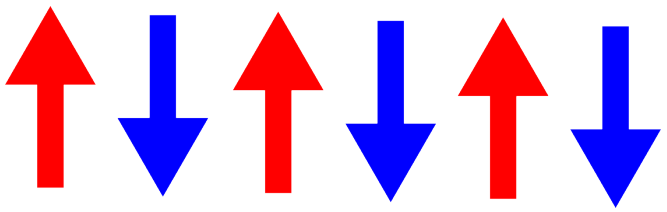
Competing solutions for a magnetic state

Let us now consider two selfconsistent solutions for the interacting model





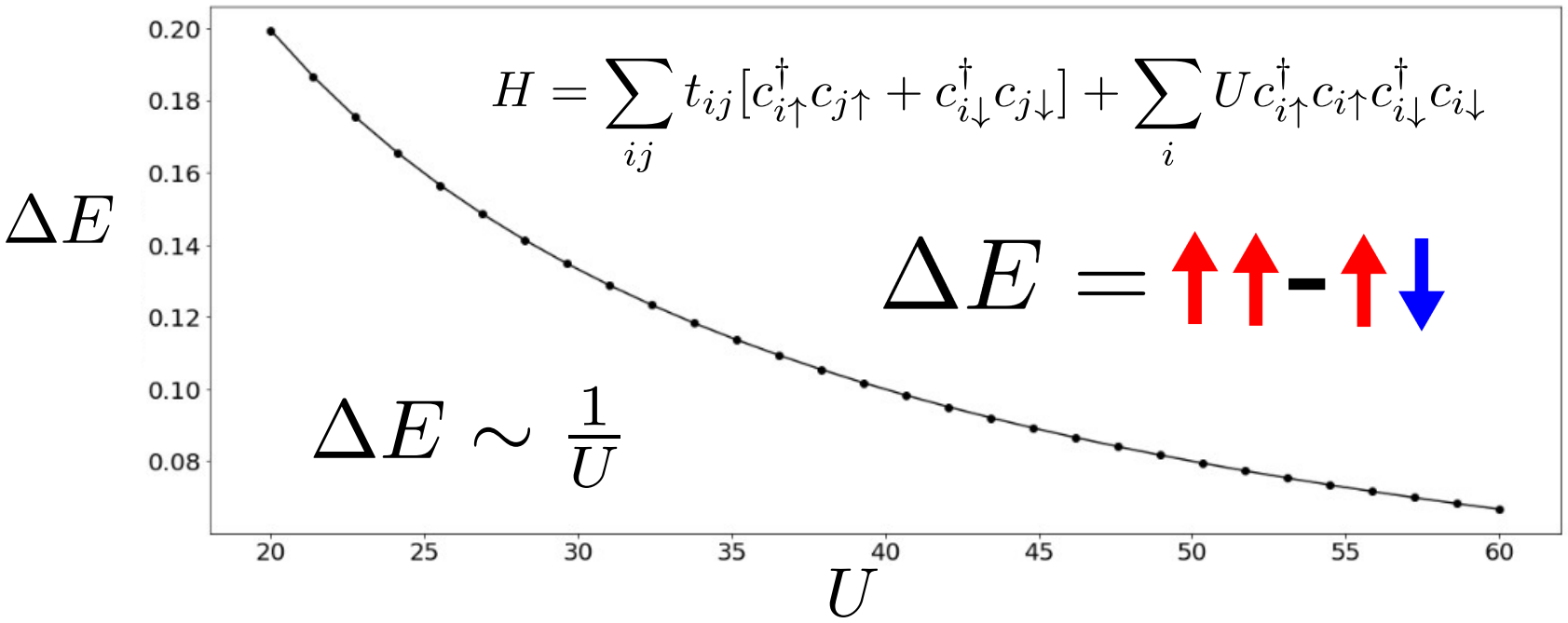




Only once of them is the true ground state, but which one it is?

Competing solutions for a magnetic state

Let us now compute the energy difference between the two configurations



For strong interactions, the AF configuration always has lower energy

The critical interaction for magnetic ordering

Lets take the Hamiltonian

$$H = \sum_{ij} t_{ij}[c^{\dagger}_{i\uparrow}c_{j\uparrow} + c^{\dagger}_{i\downarrow}c_{j\downarrow}] + \sum_{i} U c^{\dagger}_{i\uparrow}c_{i\uparrow}c^{\dagger}_{i\downarrow}c_{i\downarrow}$$

Do we have magnetism for any value of U? $\langle S_z \rangle \neq 0$

In general, in the weak coupling limit magnetism appears when

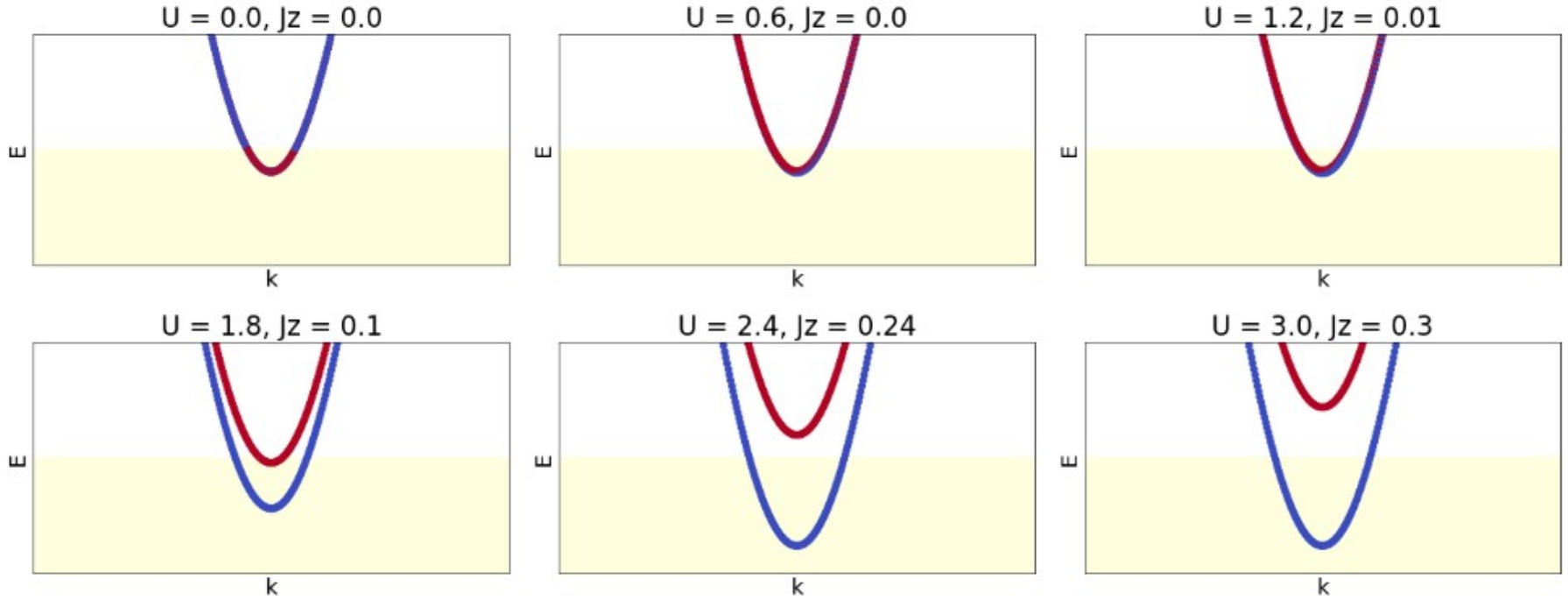
$$U D(\omega) > 1$$

Repulsive interaction

Density of states

The critical interaction for magnetic ordering

Magnetic instabilities occur once interactions are strong enough



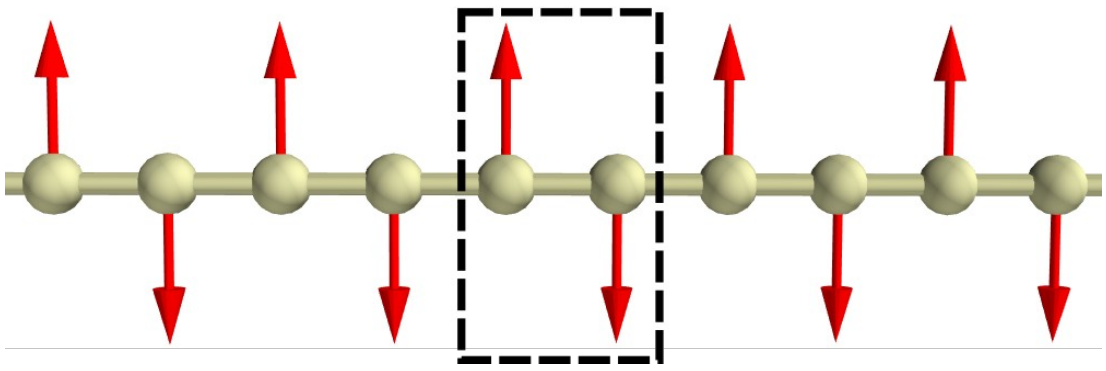
For interactions below a threshold, no magnetic order occurs

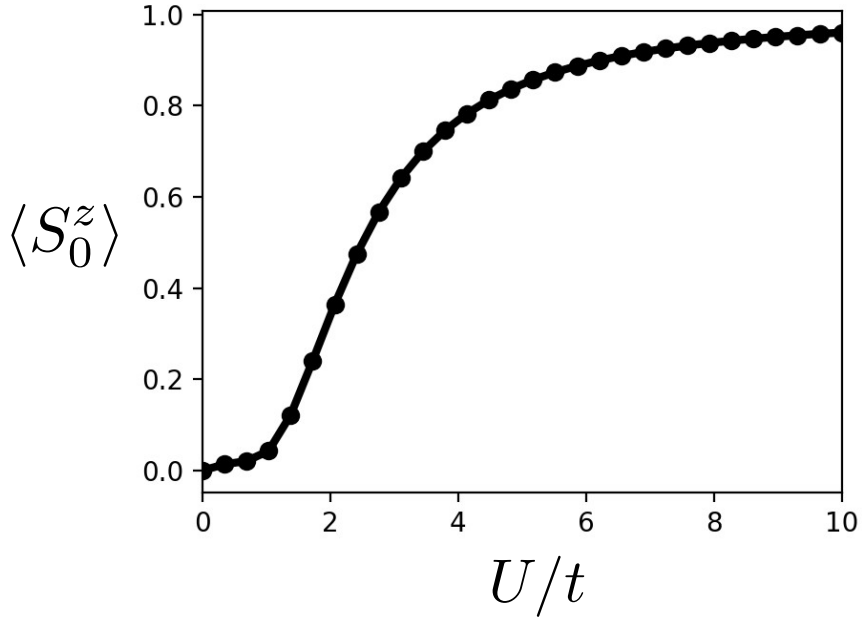
The strongly localized limit and the Heisenberg model

From a weak magnet to the strongly localized limit

$$H = \sum_{ij} t_{ij}[c^\dagger_{i\uparrow}c_{j\uparrow} + c^\dagger_{i\downarrow}c_{j\downarrow}] + \sum_i U c^\dagger_{i\uparrow}c_{i\uparrow}c^\dagger_{i\downarrow}c_{i\downarrow}$$

For large interaction strength, the system develops a local quantized magnetic moment





The strongly localized limit

Let us start with a Hubbard model dimer

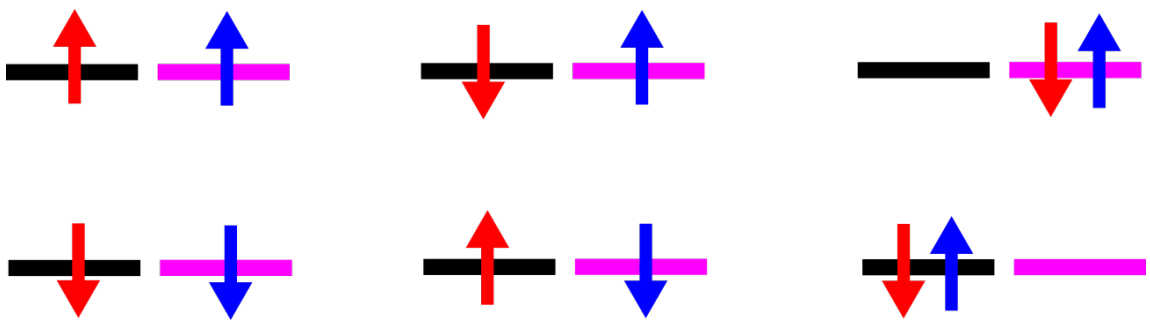
$$H = t[c^\dagger_{0\uparrow} c_{1\uparrow} + c^\dagger_{0\downarrow} c_{1\downarrow}] + \sum_i U c^\dagger_{i\uparrow} c_{i\uparrow} c^\dagger_{i\downarrow} c_{i\downarrow} + h.c.$$

Now in the limit $U \gg t$

Levels



The full Hilbert space at half filling is



The strongly localized limit

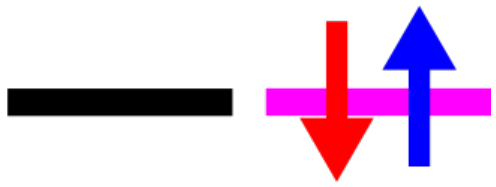
Let us start with a Hubbard model dimer

$$H = t[c^{\dagger}_{0\uparrow}c_{1\uparrow} + c^{\dagger}_{0\downarrow}c_{1\downarrow}] + \sum_i U c^{\dagger}_{i\uparrow}c_{i\uparrow}c^{\dagger}_{i\downarrow}c_{i\downarrow} + h.c.$$

The energies in the strongly localized limit are $U \gg t$



~ 0



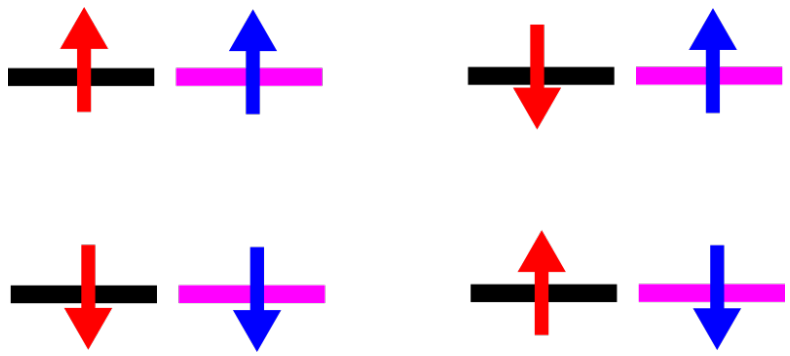
$\sim U$

The strongly localized limit

Let us start with a Hubbard model dimer

$$H = t[c^{\dagger}_{0\uparrow}c_{1\uparrow} + c^{\dagger}_{0\downarrow}c_{1\downarrow}] + \sum_i U c^{\dagger}_{i\uparrow}c_{i\uparrow}c^{\dagger}_{i\downarrow}c_{i\downarrow} + h.c.$$

The low energy manifold is



Just one electron in each site for $U \gg t$

Local S=1/2 at each site

The strongly localized limit

Effective Heisenberg model in the localized limit $\mathcal{H} = J\vec{S}_0 \cdot \vec{S}_1$

We can compute J using second order perturbation theory

$$H = H_0 + V$$

$$H_0 = \sum_i U c^\dagger_{i\uparrow} c_{i\uparrow} c^\dagger_{i\downarrow} c_{i\downarrow}$$

"pristine" Hamiltonian
(Hubbard)

$$V = t[c^\dagger_{0\uparrow} c_{1\uparrow} + c^\dagger_{0\downarrow} c_{1\downarrow}] + \text{h.c.}$$

"perturbation" Hamiltonian
(hopping)

The strongly localized limit

Effective Heisenberg model in the localized limit $\mathcal{H} = J\vec{S}_0 \cdot \vec{S}_1$

We can compute J using second order perturbation theory

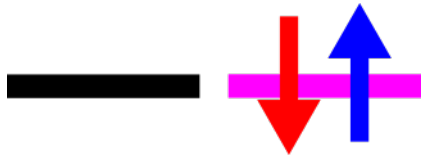
$$H = H_0 + V$$

$$J \sim \frac{t^2}{U}$$

Ground state



Virtual state



The Heisenberg model

For a generic Hamiltonian in a generic lattice

$$H = \sum_{ij} t_{ij} [c^\dagger_{i\uparrow} c_{j\uparrow} + c^\dagger_{i\downarrow} c_{j\downarrow}] + \sum_i U c^\dagger_{i\uparrow} c_{i\uparrow} c^\dagger_{i\downarrow} c_{i\downarrow}$$

In the strongly correlated (half-filled) limit we obtain a Heisenberg model

$$\mathcal{H} = \sum_{ij} J_{ij} \vec{S}_i \cdot \vec{S}_j$$

$$J_{ij} \sim \frac{|t_{ij}|^2}{U}$$

The Heisenberg model

Non-Hubbard (multiorbital) models also yield effective Heisenberg models

$$\mathcal{H} = \sum_{ij} J_{ij} \vec{S}_i \cdot \vec{S}_j$$

In those generic cases, the exchange couplings can be positive or negative

$J_{ij} > 0$

Antiferromagnetic coupling

$J_{ij} < 0$

Ferromagnetic coupling

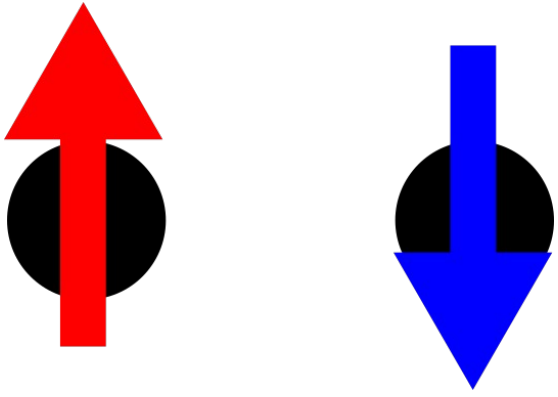
Spin-orbit coupling introduces anisotropic couplings

$$\mathcal{H} = \sum_{ij} J_{ij}^{\alpha\beta} S_i^{\alpha} S_j^{\beta}$$

The Heisenberg model

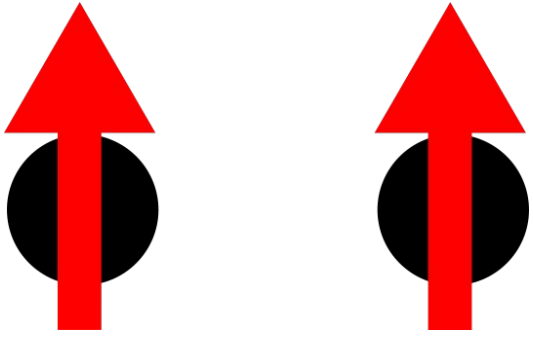
$J_{ij} > 0$

Antiferromagnetic coupling



$J_{ij} < 0$

Ferromagnetic coupling



Classical ground states

Antiferromagnetism driven by superexchange

In the square lattice



In the honeycomb lattice

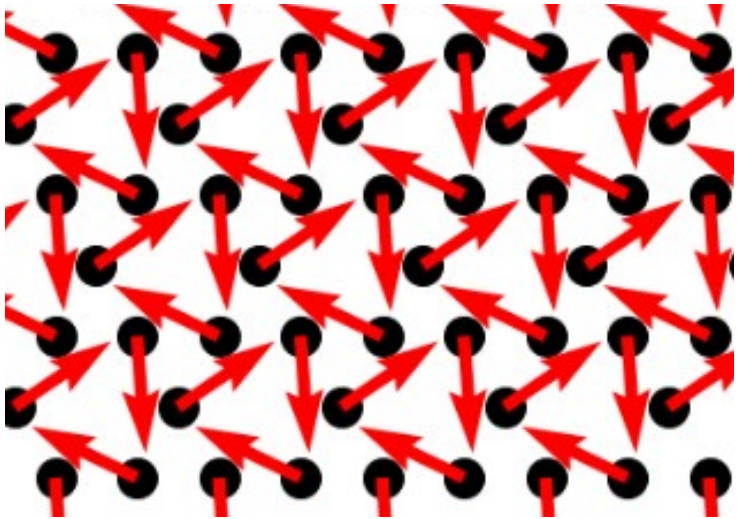


$$H = \sum_{ij,s} t_{ij} c^{\dagger}_{i,s} c_{j,s} + \sum_{i} U c^{\dagger}_{i\uparrow} c_{i\uparrow} c^{\dagger}_{i\downarrow} c_{i\downarrow} + h.c.$$

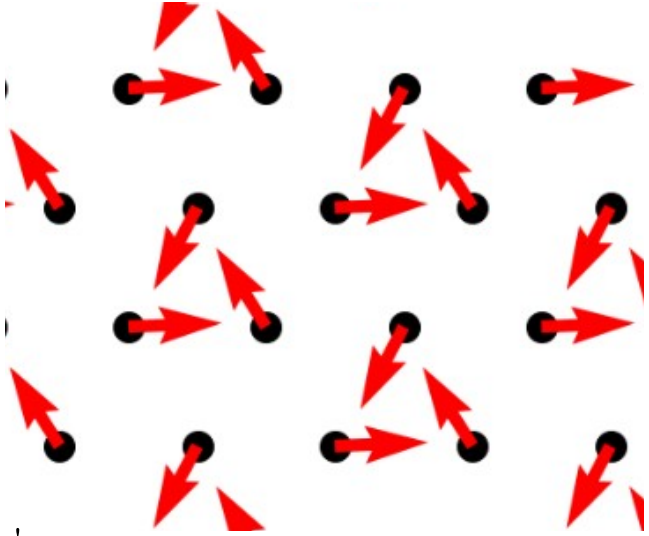
In bipartite lattices, the magnetization is collinear

Antiferromagnetism driven by superexchange

In the Kagome lattice



In the triangular lattice



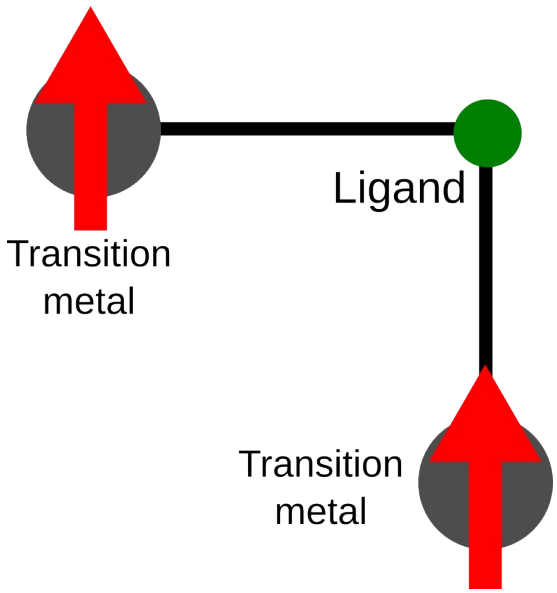
$$H = \sum_{ij,s} t_{ij} c^{\dagger}_{i,s} c_{j,s} + \sum_{i} U c^{\dagger}_{i\uparrow} c_{i\uparrow} c^{\dagger}_{i\downarrow} c_{i\downarrow} + h.c.$$

Geometric frustration promotes non-collinear order at the mean-field level

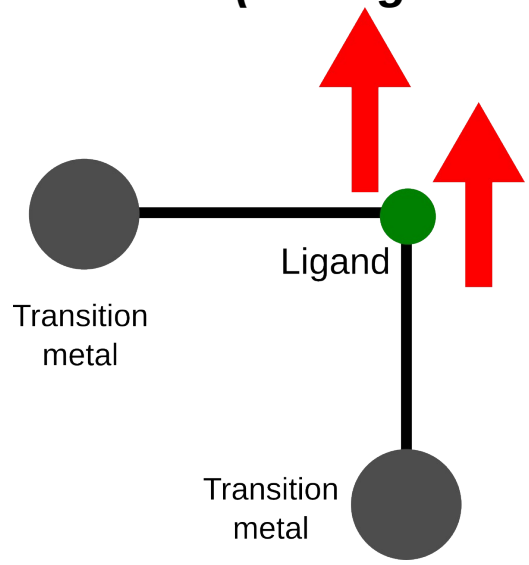
The origin of ferromagnetic coupling

Exchange interactions can be ferromagnetic if mediated by an intermediate site

Low energy manifold



Virtual state (among others)



The sign of the coupling depends on the filling of the d-shell and the angle

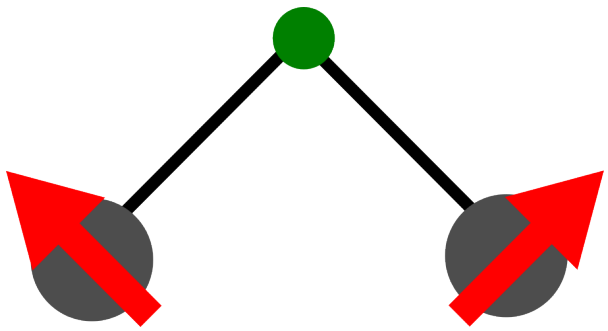
Goodenough-Kanamori rules

Non-isotropic exchange coupling

In the presence of spin-orbit coupling, new terms can appear in the Hamiltonian

Antisymmetric exchange

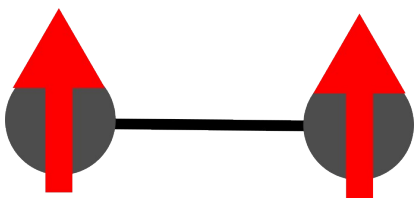
$$(\mathbf{r}_{ik} \times \mathbf{r}_{kj}) \cdot \vec{S}_i \times \vec{S}_j$$



Promotes
non-collinear order

Anisotropic exchange

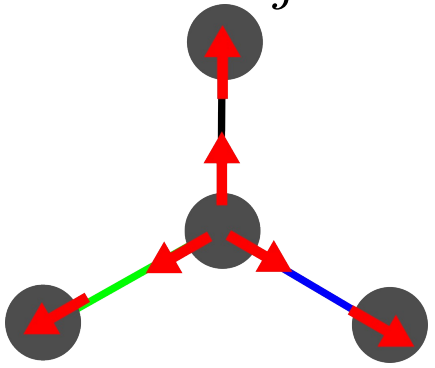
$$S_i^z S_j^z$$



Promotes
easy axis/plane

Kitaev interaction

$$S_i^{\alpha(i)} S_j^{\alpha(j)}$$

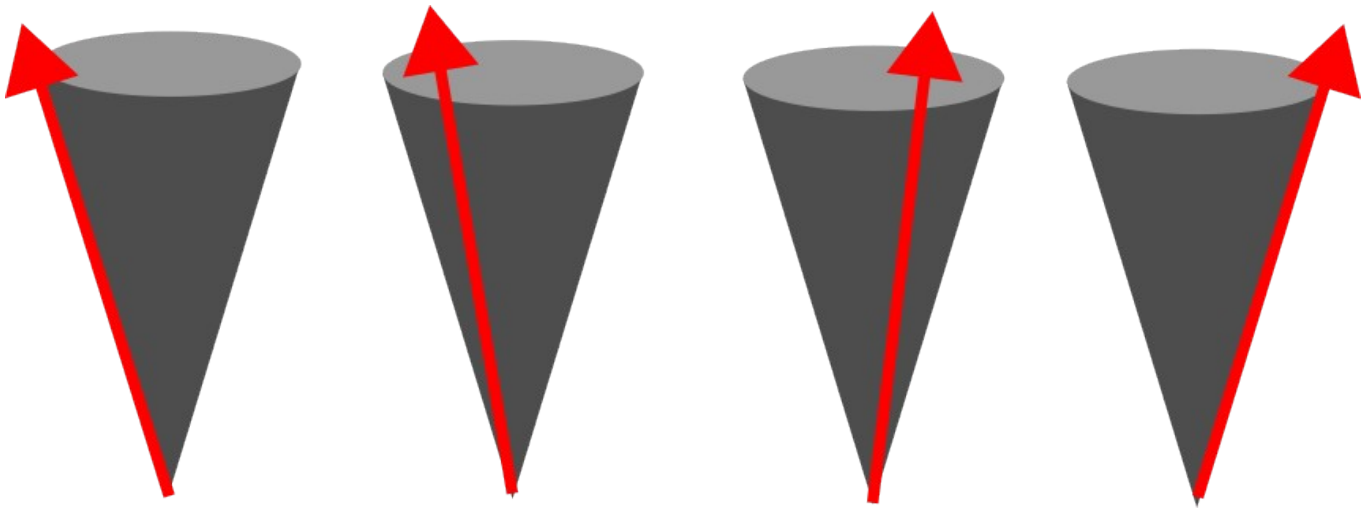


Promotes
frustration

Magnons

Excitations in a ferromagnet

Qualitatively, magnons are the fluctuations of the order parameter



Excitations in the Heisenberg model

The Heisenberg model is a full-fledged many-body problem

$$\mathcal{H} = \sum_{ij} J_{ij} \vec{S}_i \cdot \vec{S}_j$$

Algebraic commutation relations $[S_j^\alpha, S_j^\beta] = i\epsilon_{\alpha\beta\gamma} S_j^\gamma$

$$S = 1/2, 1, 3/2, 2, ...$$

How do we compute its many-body excitations?

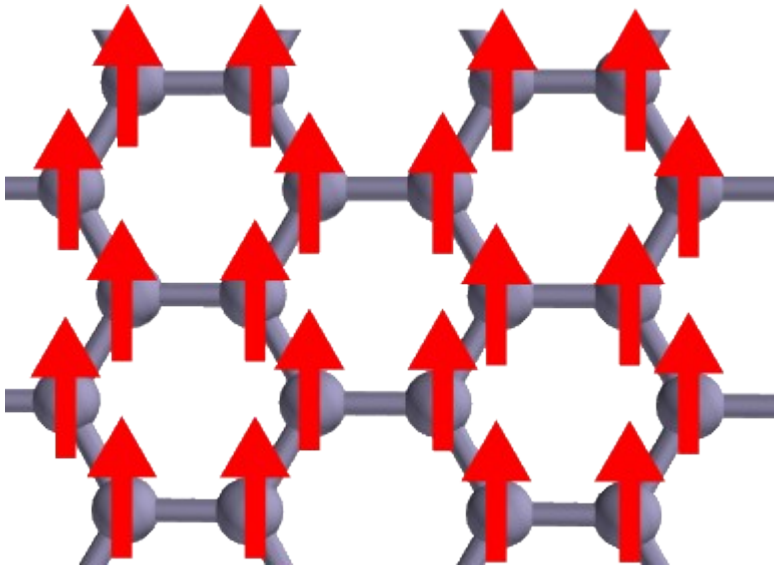
The ferromagnetic Heisenberg model

In the case of a ferromagnetic Heisenberg model, we know the ground state

$$\mathcal{H} = \sum_{ij} J_{ij} \vec{S}_i \cdot \vec{S}_j$$

$$J_{ij} < 0$$

$$|GS\rangle = |\uparrow\uparrow\uparrow\uparrow\uparrow \ldots.\rangle$$



But how do we compute the excitations?

The Holstein–Primakoff transformation

Replace the spin Hamiltonian by a bosonic Hamiltonian

$$S_+ = \sqrt{2s}\sqrt{1 - \frac{a^\dagger a}{2s}}a, \quad S_- = \sqrt{2s}a^\dagger\sqrt{1 - \frac{a^\dagger a}{2s}}, \quad S_z = \left(s - a^\dagger a\right)$$

Make the replacement and decouple with mean-field assuming $\left\langle a_i^\dagger a_i \right\rangle \ll s$

$$\mathcal{H} = \sum_{ij} J_{ij}\vec{S}_i \cdot \vec{S}_j \longrightarrow \mathcal{H} = \sum_{ij} \gamma_{ij} a_i^\dagger a_j$$

Spins **Magnon**

Magnons in a nutshell

Increase the spin $S_i^+ \sim a_i$ Destroy a magnon

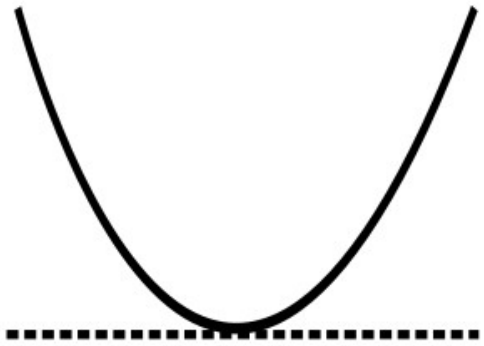
Decrease the spin $S_i^- \sim a_i^\dagger$ Create a magnon

Net magnetization $\langle S_i^z \rangle = S - \langle a_i^\dagger a_i \rangle$ Maximal minus the magnons

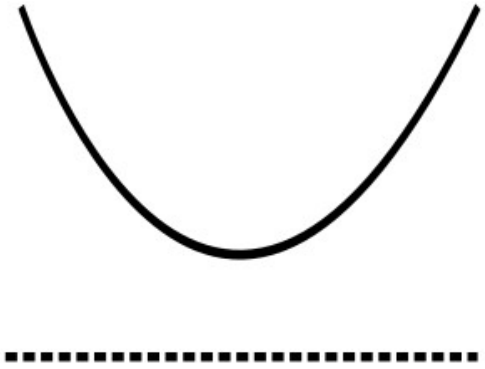
Magnons are S=1 excitations that exist over the symmetry broken state

Magnon dispersions

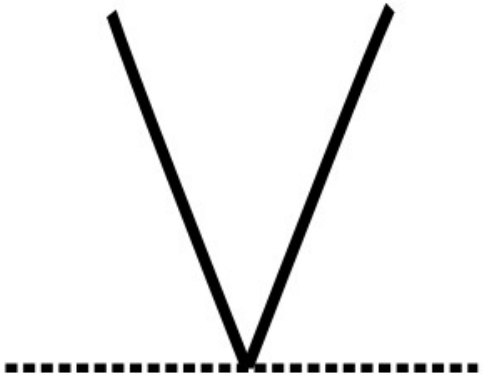
Gapless magnons



Gapped magnons



Dirac magnons

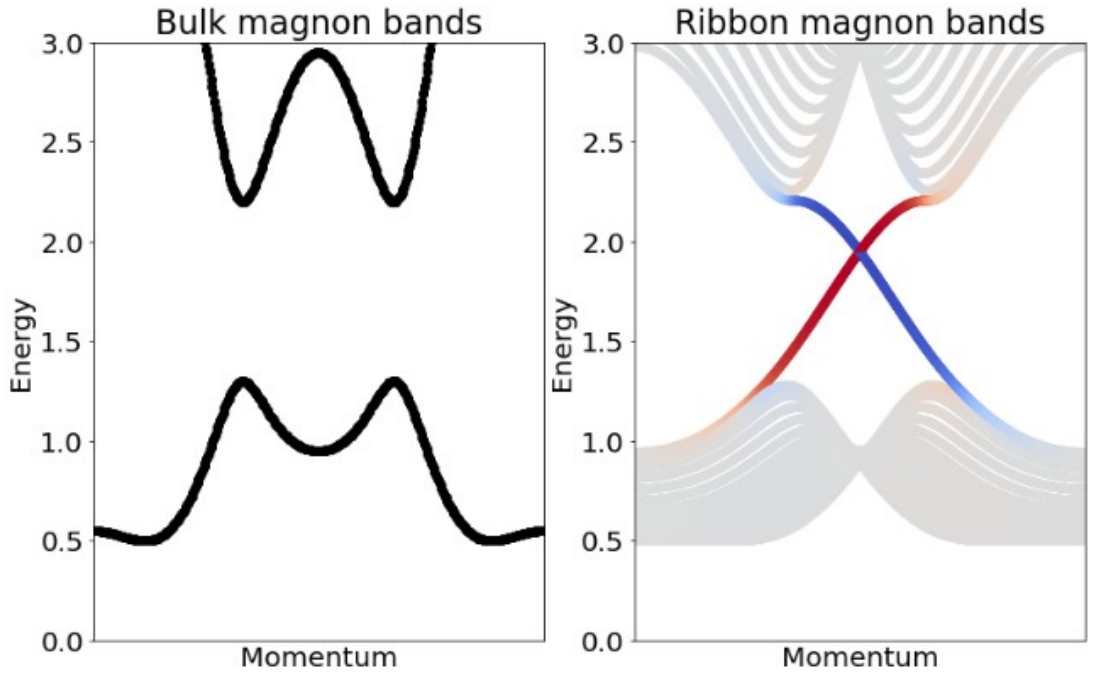


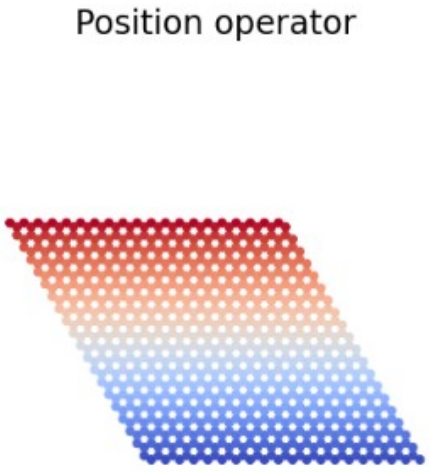
$$\mathcal{H} = \sum_{ij} \gamma_{ij} a_i^\dagger a_j$$

$$\mathcal{H} = \sum_{\mathbf{k}} \epsilon(\mathbf{k}) a_{\mathbf{k}}^\dagger a_{\mathbf{k}}$$

Topological magnons

A magnon dispersion can have topological gaps at high energies, leading to topological modes





$$\mathcal{H} = \sum_{ij} \gamma_{ij} a_i^\dagger a_j$$